

Martin Daigne

Robotics Master's Student — Data Science Minor

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Education

MSc	École Polytechnique Fédérale de Lausanne (EPFL) , Robotics - Data Science minor Core courses: Machine Learning, Deep Learning, Reinforcement Learning, Convex Optimization, Model Predictive Control, Mobile Robotics, Applied Data Analysis, Modern Natural Language Processing...	Switzerland 2024 – 2027
	Delft University of Technology (TU Delft) , Exchange - Mechanical/Aerospace Engineering	Netherlands 2023 – 2024
BSc	École Polytechnique Fédérale de Lausanne (EPFL) , Mechanical Engineering - Grade: 5.57/6 - Core courses: Analysis, General Physics (Mechanics, Thermodynamics, Electromagnetics), Numerical Analysis, Dynamical Systems, Structural Mechanics, Nonlinear Mechanics, Fluid Mechanics, Control Systems, MEMS, Autonomous Vehicles...	Switzerland 2021 – 2024

Experience

EPFL, Laboratory of Intelligent Systems (LIS) , Semester Project Design and implementation of diffusion policies (DDPM, DDIM, Score Matching, Flow Matching) to generate control trajectories for a morphing drone previously controlled with an MPPI.	Switzerland Feb 2026 – June 2026 5 months
EPFL, Computational Robot Design and Fabrication Lab (CREATE) , Semester Project Data-Enabled Model Predictive Control for Compliant Manipulators: I built a simulation in Python and an adaptive MPC controller for a soft, tendon-driven arm using CasADi and acados for real-time deployment.	Switzerland Sept 2025 – Jan 2026 5 months
EPFL, Unsteady Flow Diagnostics Laboratory (UNFoLD) , Summer in the Lab Internship Design, building, and analysis of a random gust generator to improve realism in wind turbine experiments.	Switzerland June 2024 – Aug 2024 3 months
Aether Swiss Kite , Control Team Leader Lead the data acquisition, system identification, and control of the kite to achieve optimized automated control.	Switzerland 2025 – 2026 1 year
Aether Swiss Kite , Electrical and Control Engineer 1st semester: Design, building, and integration of a position sensor of the kite. 2nd semester: Data acquisition and filter implementation (EKF, UKF) to localize the kite.	Switzerland 2024 – 2025 1 year
EPFL Rocket Team , Recovery System Engineer Design and testing of the parachute of the Nordend rocket of the EPFL Rocket Team for the European EUROCC competition.	Switzerland 2022 – 2023 1 year
EPFL , Teaching Assistant Teaching assistant in Introduction to Computer Science, Analysis II, Fluid Mechanics, and Model Predictive Control.	Switzerland 2022–2023 / 2024–2025

Awards

Deep Learning Object Detection Competition

2025

Winner of the Deep Learning Object Detection competition 2025 in the Image Analysis and Pattern Recognition course at EPFL.

EPFL Space Race

2022

Winner EPFL Space Race 2022.

Skills

Coding Languages: Strong experience in Python (PyTorch, Pandas, scikit-learn, ...). Worked on Matlab and knowledge in C/C++ (mostly for Arduino).

Languages

French: Mother tongue

English: C1 (Cambridge English Advanced Certificate, 2020)

Dutch: Fluent

German: Scholar (Goeth Institut, B1, 2018)